



Venkata Sai Kowshik Gowrakkavari

India

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Education

Bachelor's of Technology in
Computer Science and
Engineering, Amrita Vishwa
Vidyapeetham, IN, 2025

Digital Credentials

AWS Partner: Generative AI
Essentials - Training Badge -
2026

Enterprise Design Thinking
Practitioner - 2025

IBM Generative AI Foundations -
2025

IBM Generative & Agentic AI
Foundation - 2025

IBM Garage Essentials - 2025

Languages

English - Fluent

Telugu - Fluent

Hindi - Basic

Publications

Enhancing Patient-AI Interaction
in Healthcare using Sentiment
Analysis with BERT and Hybrid
Deep Learning Models, 2025

Profile

I am a proactive and dedicated Computer Science student with a CGPA of 8.56 from Amrita Vishwa Vidyapeetham. My technical skills span programming languages such as Java, C++, Python, and JavaScript, along with web technologies including ReactJS, NodeJS, HTML, and CSS. I am experienced in working with databases like PostgreSQL and MongoDB, and familiar with development tools such as Visual Studio Code and Eclipse.

I have applied these skills in impactful projects, including MediVault (a secure medical data platform with AI-driven insights and AES encryption), Nutri-Check (a nutrition tracking application integrated with Google Fit), and a University Admission Prediction System (achieving 98% accuracy using Gradient Boosting and Neural Networks). I also developed a House Boat Management System to optimize operations with secure authentication and efficient data handling.

Beyond academics, I have actively contributed to community initiatives, most notably leading Roots Reimagined, an outreach program that delivered culturally enriching education to rural students.

Key skills

• Programming & Software Development

- Proficient in Java, C++, Python, and JavaScript
- Strong knowledge of OOPS, data structures, and algorithms

• Web Development

- Full-stack development with ReactJS, NodeJS, HTML, and CSS
- Experience with API's

• Machine Learning & AI

- Worked with Gradient Boosting and Neural Networks in academic and practical projects
- Applied ML models for prediction and health insights integration

• Cloud Technologies

- AWS
- Google Cloud

• Cybersecurity & Data Privacy

- Implemented AES encryption, dynamic access codes, and authentication protocols
- Basic understanding of data protection standards

• Database & Backend

- Experience with PostgreSQL and MongoDB
- Integration using JDBC and Mongoose; schema design for real-time applications

• Cloud & DevOps

- Familiar with cloud deployments and CI/CD practices
- Version control and collaboration using Git/GitHub



- **Project Management & Collaboration**

- Contributed to hackathons and outreach programs
- Knowledge of Agile practices including sprint planning and retrospectives

Key courses and training

GCP Associate Cloud Engineer - Google Cloud Certification

Work experience

Personal Project

-, India | Apr 2025 - Jun 2025

QuickPixel

- Built a **Streamlit-based web application** for generating AI-powered product advertisements using **Bria API**.
- Implemented backend workflows for **prompt enhancement, background removal, shadow effects, and CTA overlays**.
- Integrated **image generation, customization, and instant download features** for smooth user experience.

Tools & Tech: Python, Streamlit, Bria API, Image Processing, REST APIs

Personal Project

-, India | Aug 2024 - Oct 2024

Edu-Aid

- Developed **backend pipelines** for an AI-powered academic assistant, answering complex queries in real-time.
- Designed **data retrieval logic** to support last-minute exam preparation and knowledge reinforcement.
- Scaled backend workflows to handle **concurrent student queries** with low latency.

Tools & Tech: Python, NLP, AI Model Integration, Backend Architecture

Major Project

-, India | Jul 2024 - Dec 2024

MediVault

- Contributed to a secure medical data management and telemedicine platform with AI-driven analytics.
- Developed backend modules for AES-256/RSA encryption, Aadhaar-based OTP authentication, and role-based access control.
- Built ML pipelines for personalized health recommendations and predictive health risk analysis.
- Integrated real-time chat communication (Socket.IO) and supported scalable data storage with MongoDB.

Role: Backend & AI Development

Tools & Tech: Python, Flask, MongoDB, PostgreSQL, TensorFlow, Scikit-learn, NLP, Socket.IO

Personal Project



-, India | Mar 2024 - May 2024

University Admission Prediction System

- Designed a **data-driven ML system** to predict university admissions using academic and extracurricular features.
 - Evaluated multiple algorithms (**Decision Tree, SVM, Naive Bayes, Gradient Boosting, Neural Networks**) on **10,000+ records**.
 - Achieved **98% accuracy with Gradient Boosting**, identifying **GRE, CGPA, and research experience** as top predictors.
 - Built pipelines for **training, testing, and visualization** of model performance.
- Tools & Tech:** Python, Scikit-learn, TensorFlow, Pandas, Matplotlib, Seaborn

Personal Project

-, India | Feb 2022 - Apr 2022

Bhagavad Gita Geeks

- Built an **online e-reading platform** enabling users to explore any chapter of the Bhagavad Gita.
 - Designed backend logic for **chapter retrieval, rendering, and seamless navigation**.
 - Focused on **scalability and accessibility** to ensure uninterrupted engagement across devices.
- Tools & Tech:** Django/Flask (Python), HTML, CSS, JavaScript, Database and API's Integration.

Personal Project

-, India | Feb 2022 - Jun 2022

House Boat Management System

- Built a **Java desktop application backend** to manage bookings, authentication, and secure data workflows.
 - Integrated **PostgreSQL database with JDBC** for reliable data persistence.
 - Designed booking workflows for **owners and employees**, enabling real-time updates.
- Tools & Tech:** Java, Swing, JDBC, PostgreSQL

Assignment history

Application Developer

canara, India | Nov 2025 - Present

Project name: Project Tejas(canfinhomes)

CanFin initiated a transformation program aimed at enhancing customer experience, operational efficiency, and revenue growth. The primary objective of the project was to supply, implement, and maintain the Core Business Solution infrastructure and security solutions on a public and/or private cloud platform. The term "Core Business Solution (CBS)" encompasses the Lending Solution (including LOS, LMS, and Collections), Document Management System (DMS), Accounting, Borrowing and Investment modules, Deposits, Risk Management and ALM, Aadhaar Data Vault, HRMS, reporting capabilities, and the underlying cloud-based IT infrastructure.



Contribution

1. Processing & Pushing Multiple CSV Files to DB (API-Based Automation)

- Developed automation to **process and push multiple CSV files into database systems via APIs**.
- Built logic to:
 - Read and parse **bulk CSV files dynamically**
 - Convert data into API-compatible payloads
 - Trigger API calls for database insertion
- Implemented **batch-wise processing** to handle large datasets efficiently.
- Added **error handling and response validation** to track failed API transactions.
- Designed file handling workflow to:
 - Process files sequentially
 - Maintain logs for processed records
- Improved consistency and reduced manual data upload efforts through automation.

2. Enterprise Active Directory User Automation

- Developed **automation scripts (PowerShell/Python)** for enterprise-level **AD user lifecycle management**.
- Built solution to:
 - Automate **bulk user creation, updates, and group assignments**
 - Process structured input files (CSV/Excel)
- Implemented:
 - Validation checks before user provisioning
 - Logging for tracking success/failure of operations
- Ensured **standardized user creation aligned with organizational policies**.
- Integrated **automated email notifications**, where:

Emails are triggered to respective recipients defined in the input CSV

Users receive communication upon successful account creation or updates

- Reduced dependency on manual AD operations by introducing repeatable automation workflows.

3. Large-Scale Data Migration (High Volume – ~1 Crore Records)

- Engineered data migration workflows to handle **high-volume datasets (~1 Crore records)** from source files to target systems.
- Developed logic to:
 - **Split records based on log validation** into:
 - Valid records → processed for migration
 - Invalid records → redirected for error handling
- Performed:
 - Source vs target **data validation and reconciliation**
 - Record-level verification to ensure consistency
- Built scripts to:
 - Process data in **chunks/batches** for stability
 - Generate **migration status reports**
- Improved migration reliability by ensuring only **validated data is pushed to the database**.

4. Active–Passive Node Automation (AWS EC2)

- Worked on automation for **active-passive node setup using AWS EC2 instances**.
- Contributed to:
 - Environment setup and configuration for multiple nodes



- Supporting failover-ready architecture
- Assisted in automating:
 - Node readiness checks
 - Basic deployment consistency across instances
- Helped improve system reliability by enabling **structured node management and failover support**.

5. AWS ETL Pipeline Development

- Designed and implemented **end-to-end ETL pipelines using AWS Glue**:
 - Source → **Amazon S3 (CSV/Excel)**
 - Transformation → **AWS Glue (PySpark/Pandas)**
 - Target → **PostgreSQL staging DB**
 - Built automation to:
 - Process **multiple files sequentially from S3**
 - Move files into **processed/error folders** based on outcome
 - Developed transformation logic to:
 - Handle **schema mismatches and extra columns**
 - Clean and validate data before loading
 - Integrated **AWS APIs and services** for:
 - Data access and movement
 - Secure execution using IAM roles
 - Implemented:
 - Logging via **CloudWatch**
 - Error handling for failed transformations
 - Achieved processing performance of **~12,000 records within ~1.5–2.5 minutes**.
- ### Technical Focus & Strengths
- Strong inclination towards **AWS-based development (Glue, S3, EC2, IAM, CloudWatch)**
 - Hands-on experience in **automation using Python & PowerShell**
 - Experience working with **APIs for data ingestion and processing**
 - Focused on **reducing manual effort through scripting and workflow automation**
 - Built solutions emphasizing **data validation, reliability, and scalability**

Additional information

Successfully led and contributed to winning teams in multiple internal and external hackathons, demonstrating innovation, collaboration, and rapid problem-solving.

Hackathon Achievements

- HackCEC 2023: Blockchain-based coupon trading app
- Pegasus 2.0: EduAid – Generative AI for lecture summarization
- GeeksForGeeks Hackathon: Nutri-Check – Nutrition tracker with Google Fit

Community Leadership

Led Roots Reimagined 2023 – a cultural education initiative for rural students. Promoted tech-driven social impact.